Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Access the Proxmox Hypervisor and understand some of the features
- Deploy a virtual machine with an existing VM image
- Access the virtual machine via remote console
- Connect the virtual machine to the network switch

Required Hardware

- 1 x Server with IPMI port
- 1 x RJ45 Ethernet Cable
- 1 x Laptop

Required Software

- Web Browser

PART 1: Accessing the Proxmox Hypervisor and Understanding Some of the Features

In this lab, you are going to learn the basic features of the Proxmox VM (virtual machine) hypervisor.

Referring to the previous lab, "Getting to Know the Server" Part 4, you should be able to access the Proxmox dashboard using the following steps:

1.1 Proxmox Hypervisor is accessible from the CLI console and the web management dashboard for easy management of virtual machines. To check the access to the Proxmox web dashboard, connect your laptop to port "LAN 1" using an RJ45 Ethernet cable, as shown in Figure 1.1 below.



Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.1

After it is connected, you can access the Proxmox web management dashboard by opening this URL in your web browser: https://10.0.0.10:8006
If a privacy error page appears, expand "advanced" and click on "Proceed to 10.0.0.10" to continue.

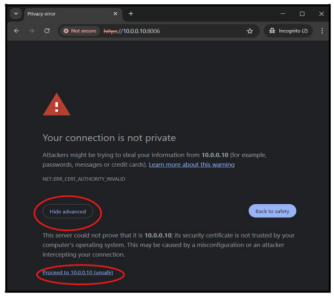


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.2

1.2 If it is successful, you will see the Proxmox Web Dashboard.

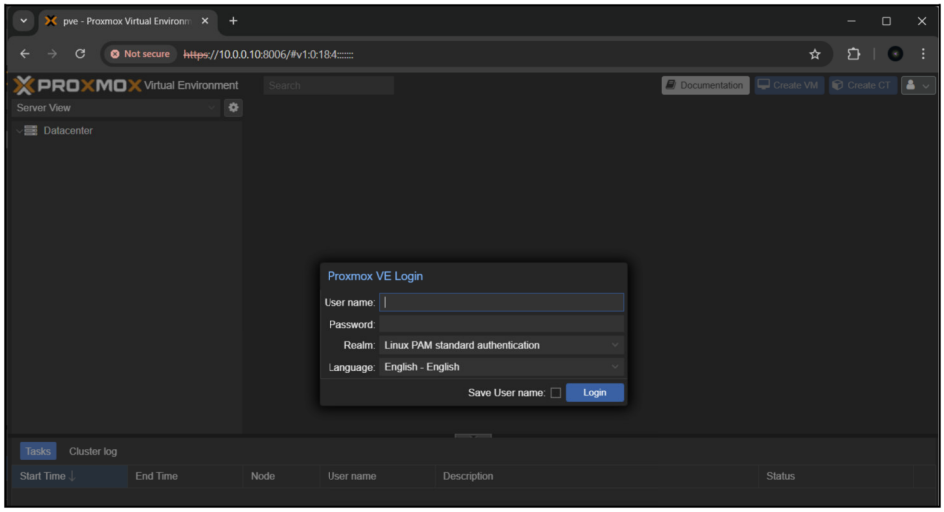


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.3

1.3 Log in to the Proxmox Web Dashboard using the **root** account with the password written on the whiteboard to continue.

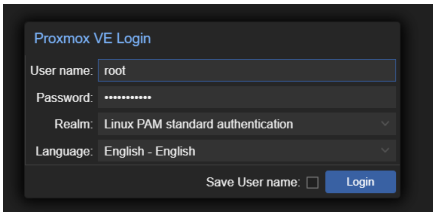


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.4

1.4 If the login is successful, you will see the Proxmox Dashboard as shown below.

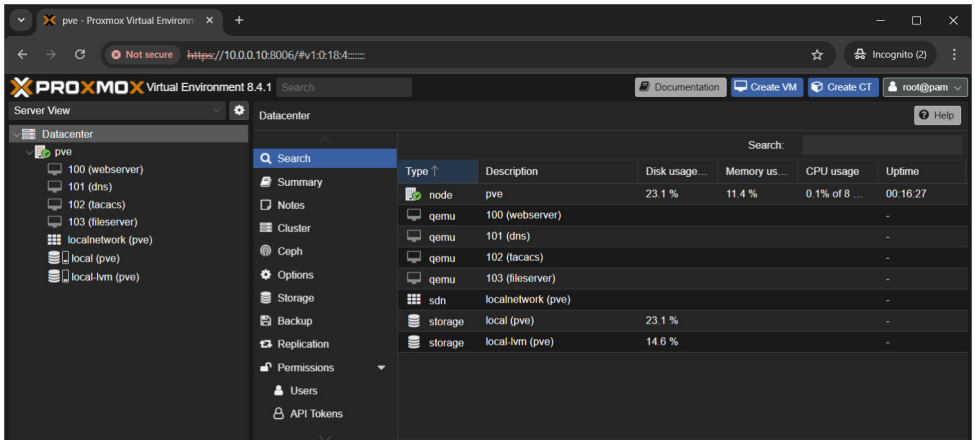


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.5

1.5 The most important information of the Proxmox Hypervisor is the capacity of the physical resources that the hypervisor can manage (e.g., CPU, RAM, Hard disk, Network Interface, and others). Click on the “pve” and then “Summary” to check on the capacity and status of the hardware specification.

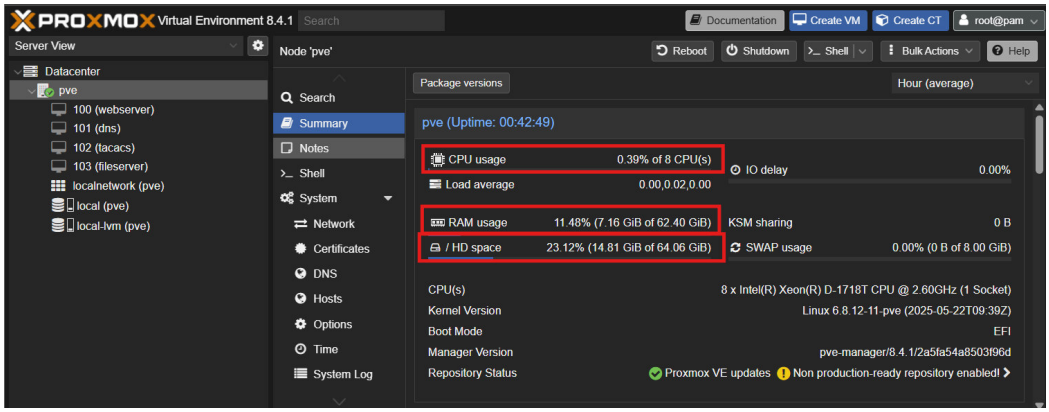


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.6

1.6 **For the network interface**, the list of physical interfaces can be shown by clicking on “pve”, followed by “System” and “Network” to see the list of default networks configured in the Proxmox server.

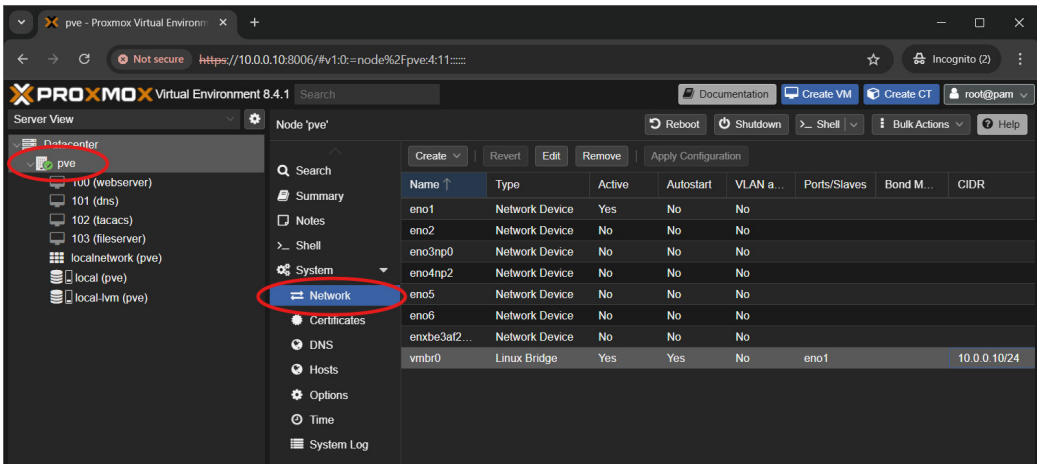


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.7

No.	Name	Type	Descriptions
1	eno1	Network Device	Physical Port – LAN 1
2	eno2	Network Device	Physical Port – LAN 2
3	eno3np0	Network Device	Physical Port – SFP port 1
4	eno4np2	Network Device	Physical Port – SFP port 2
5	eno5	Network Device	Physical Port – LAN 3
6	eno6	Network Device	Physical Port – LAN 4
7	enxbe3af2b6059f	Network Device	Physical Port – IPMI
8	vmbr0	Linux Bridge	Mapped to port eno1, assigned IP 10.0.0.10/24 for Proxmox hypervisor access

1.7 **For the storage capacity**, the list of physical hard disks can be shown by clicking on “local (pve)” or “local-lvm (pve)” to check the available storage.

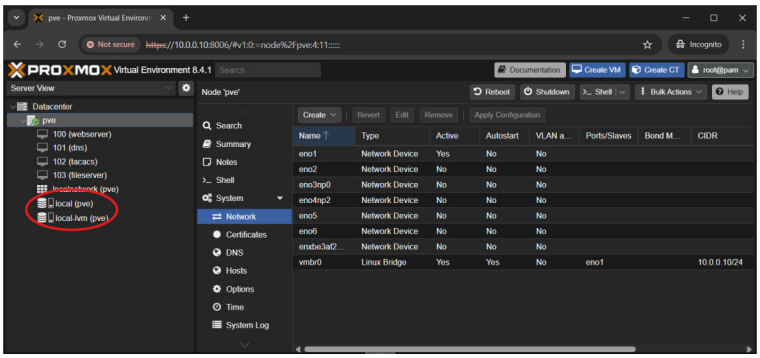


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.8

Local (pve) is a physical disk partition that is mainly used to store raw data, including uploading of ISO images and backup data.

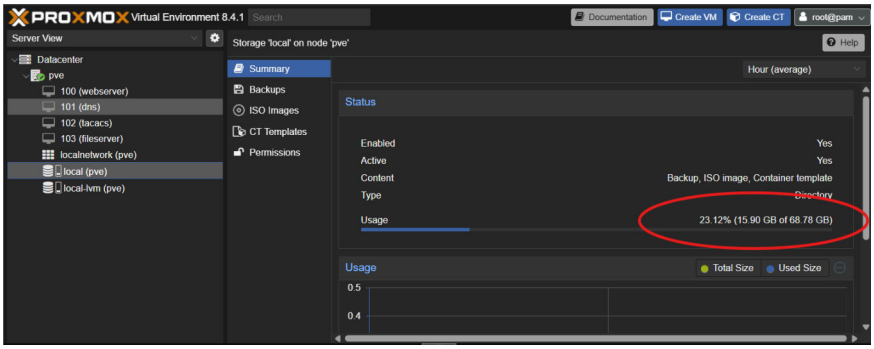


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.9

Local-lvm (pve) is a logical partition that is mainly to store or keep the Virtual Machine (VM) disks and container (CT) volumes.

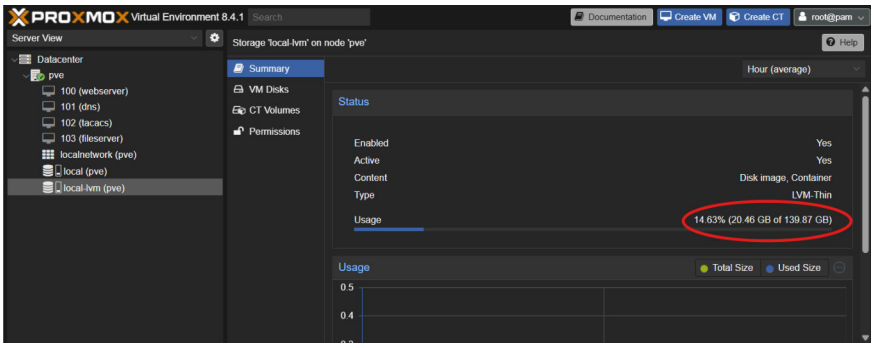


Figure Accessing the Proxmox Hypervisor and Understanding Some of the Features.10

PART 2: Start and Access an Existing Virtual Machine

In this part, you are going to learn how to start and access the pre-configured virtual machine based on your requirements. By default, there are 4 (four) different pre-configured Linux-based virtual machines with different packages.

2.1 There are four Virtual Machines (VMs) Pre-configured in this Proxmox Server for use. You may use these VMs for your project or learning use cases.

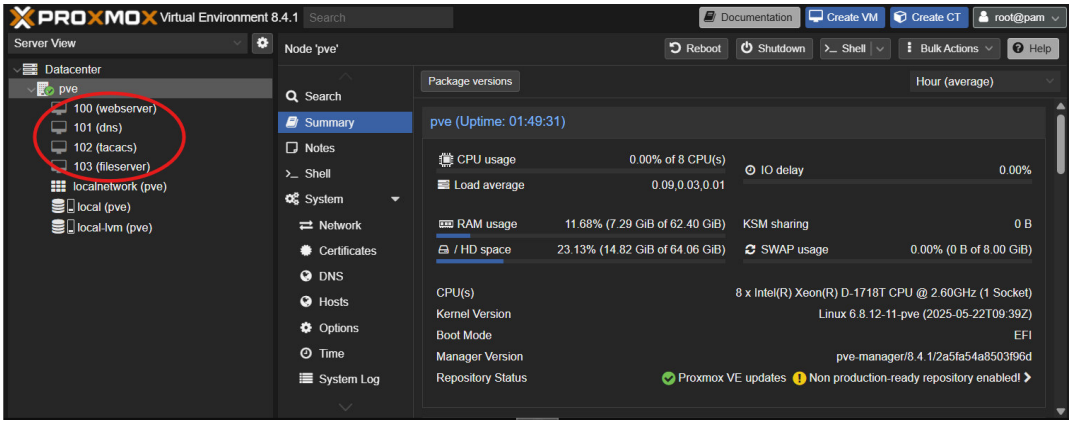


Figure Start and Access an Existing Virtual Machine .11

Name	Packages	Functions	Preconfigured IP
100 (Webserver)	Apache HTTP2	Running a web server	10.0.0.11/24
101 (DNS)	Bind9	Provide a DNS resolver or server	10.0.0.12/24
102 (Tacacs)	Free Radius + TACACS	Provide AAA service	10.0.0.13/24
104 (File Server)	FTP + SMB	Provide file transfer/sharing service	10.0.0.14/24

Gateway configured for all VMs: 10.0.0.254
DNS configured for all VMs: 8.8.8.8

Starting a Virtual Machine

2.2 To start a Virtual Machine (VM) that is powered off, click on the desired VM name, under the Summary tab, click on "Start" button on the top right.

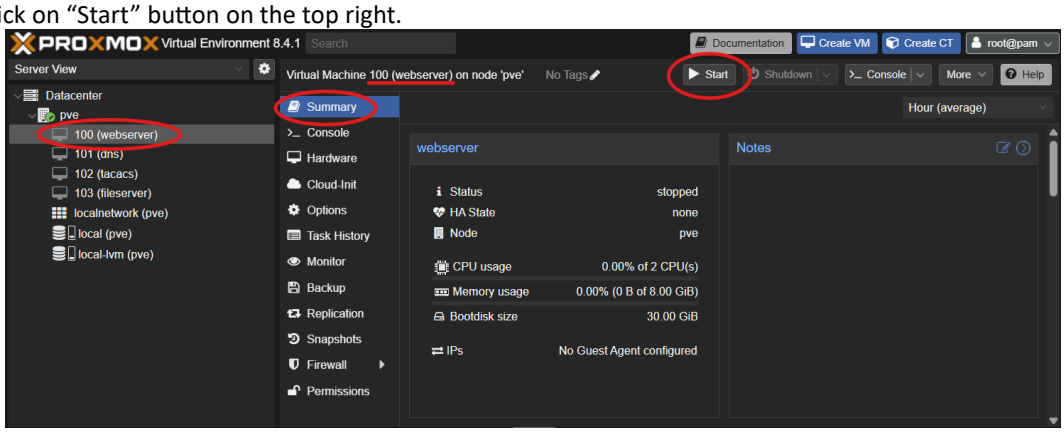


Figure Start and Access an Existing Virtual Machine .12

2.3 Notice the Summary status of the VM had changed. You have successfully powered on a virtual machine.

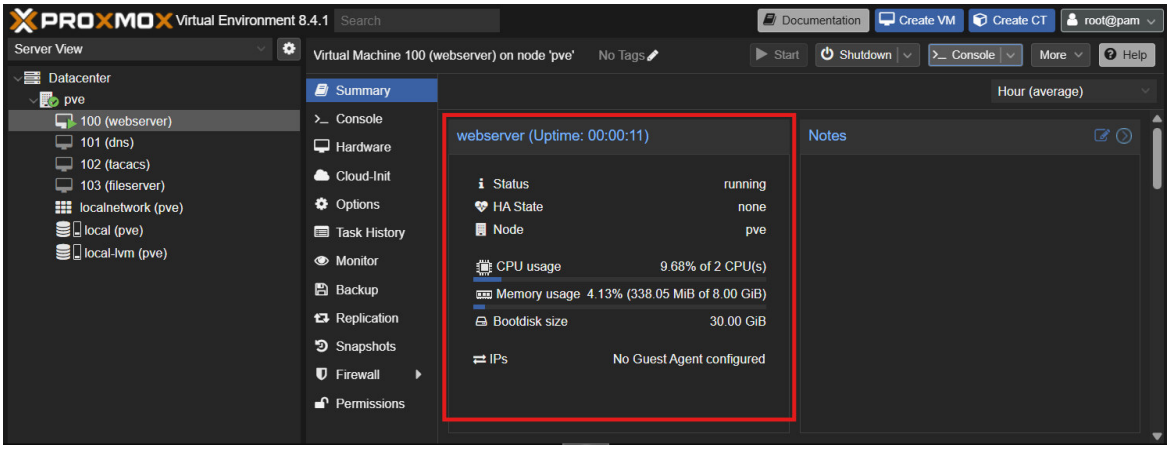


Figure Start and Access an Existing Virtual Machine .13

Access the Virtual Machine via Remote Console

In general, a virtual machine has capabilities and features similar to those of a common PC or server, where it needs to have a physical display or monitor to manage configuration at the operating system (OS) level. However, VM doesn't need a physical display or monitor to change configuration at the OS level, because Proxmox has a remote console to access every individual VM.

2.4 To access a VM via remote console, click on any powered-on VM in the Proxmox Dashboard, and click on the **">_ Console"** tab.

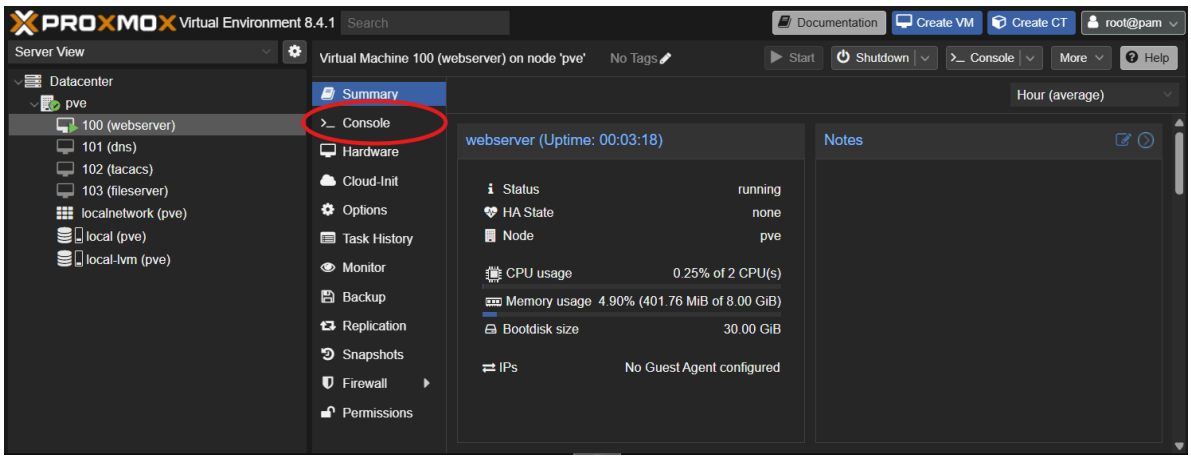


Figure Start and Access an Existing Virtual Machine .14

2.5 On the **">_Console"** tab, you will see a minimized version of the virtual machine console connection. To expand for a bigger console screen, you can either click on the top right **">_Console"** button again for a separate pop-out window console connection or click the full screen button located beside the minimized console.

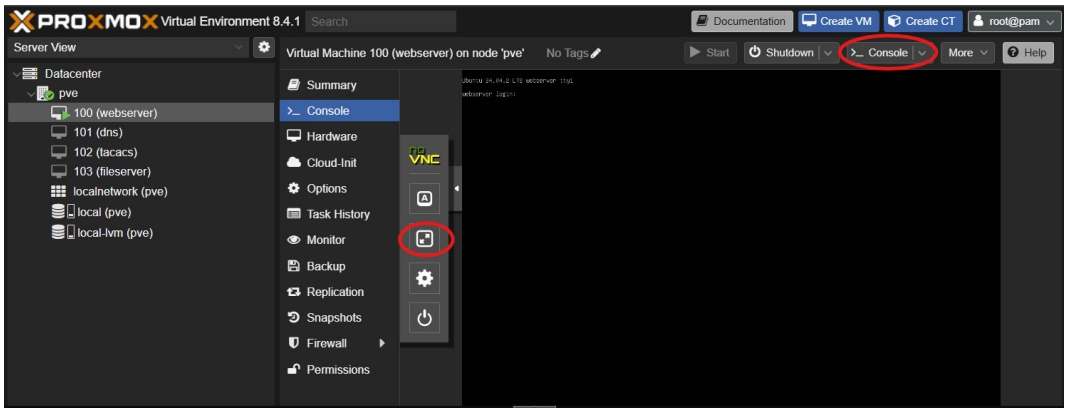


Figure Start and Access an Existing Virtual Machine .15

2.6 To log in to any of the four preconfigured virtual machines via the console, the default username is **ubuntu** password as mentioned in the whiteboard.

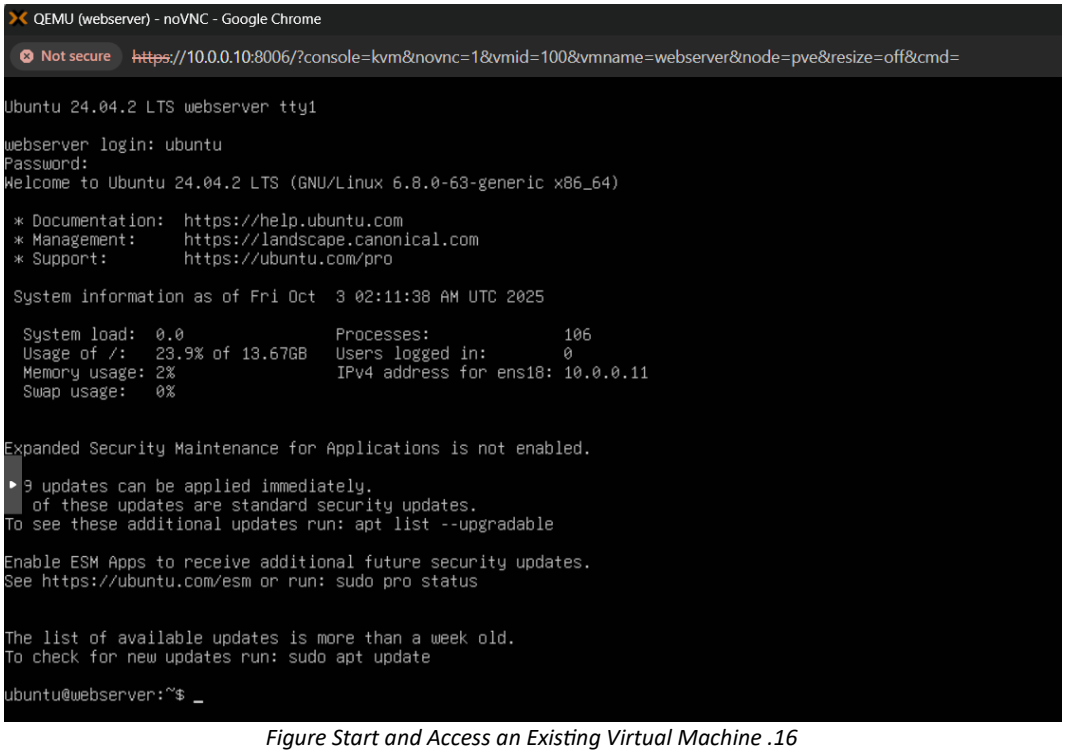


Figure Start and Access an Existing Virtual Machine .16

PART 3: Connect the Virtual Machine to the Network Switch

By default, the virtual machine only has a virtual network interface to connect to other virtual machines. A virtual machine can never connect to a network device (LAN port) directly because it can only be connected to a Linux bridge located in the network configuration of the Proxmox hypervisor.

3.1 To check the full list of network types available, click on the **pve → System → Network**

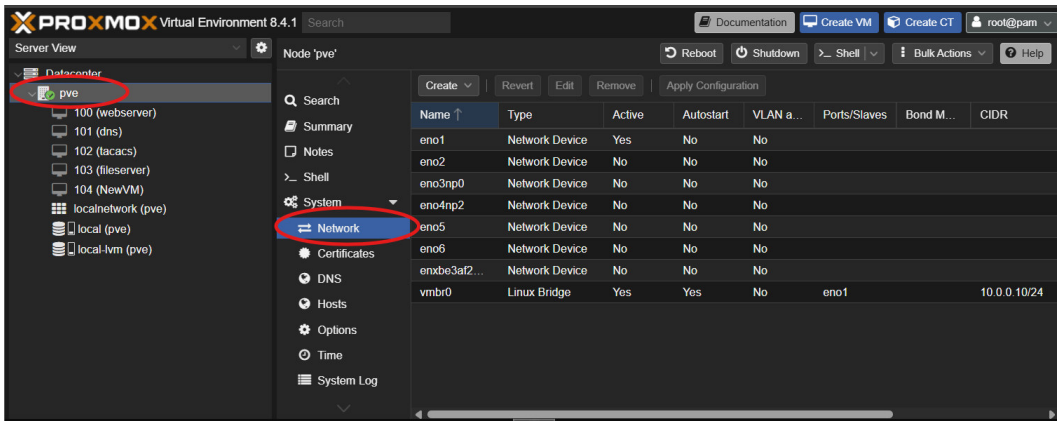


Figure Connect the Virtual Machine to the Network Switch.17

3.2 To create a new Linux bridge in Proxmox Hypervisor, go to the network page, and click on **"Create"**

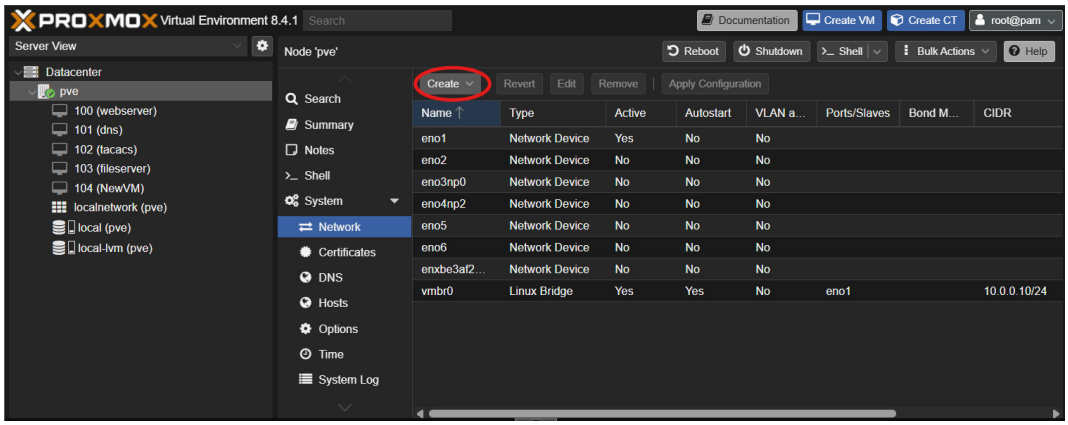


Figure Connect the Virtual Machine to the Network Switch.18

3.3 Under the dropdown button, choose **"Linux Bridge"**.

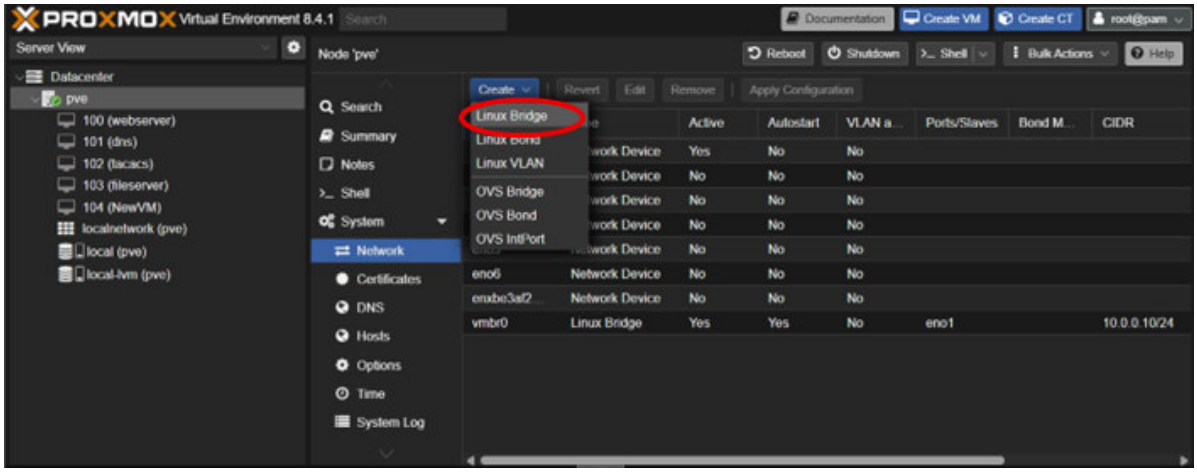


Figure Connect the Virtual Machine to the Network Switch.19

3.4 In the "**Create: Linux Bridge"** prompt, enter any name for the bridge (for example, **vmbr1**) and key in the respective network device name (for example, **eno2**) into the bridge, and then click **Create**.

*Note: You may need to create multiple bridges with different respective network device names (**eno3** and **eno4**) for different subnets of your network design (e.g., **LAN, DMZ, OOB**)*

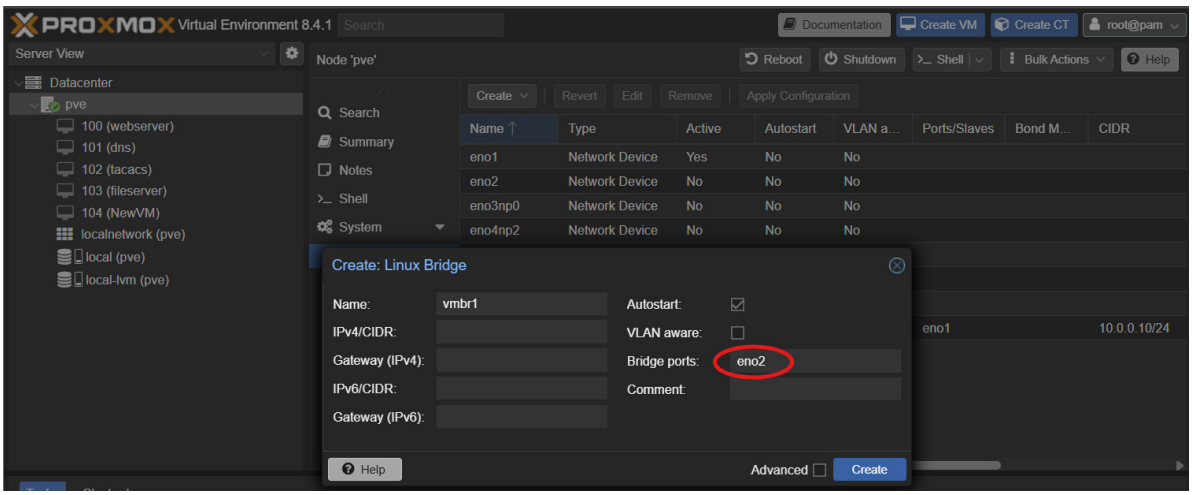


Figure Connect the Virtual Machine to the Network Switch.20

3.5 Once created, it should appear in the network list, and then click **"Apply Configuration**" for the Linux bridge to be created and ready to use.

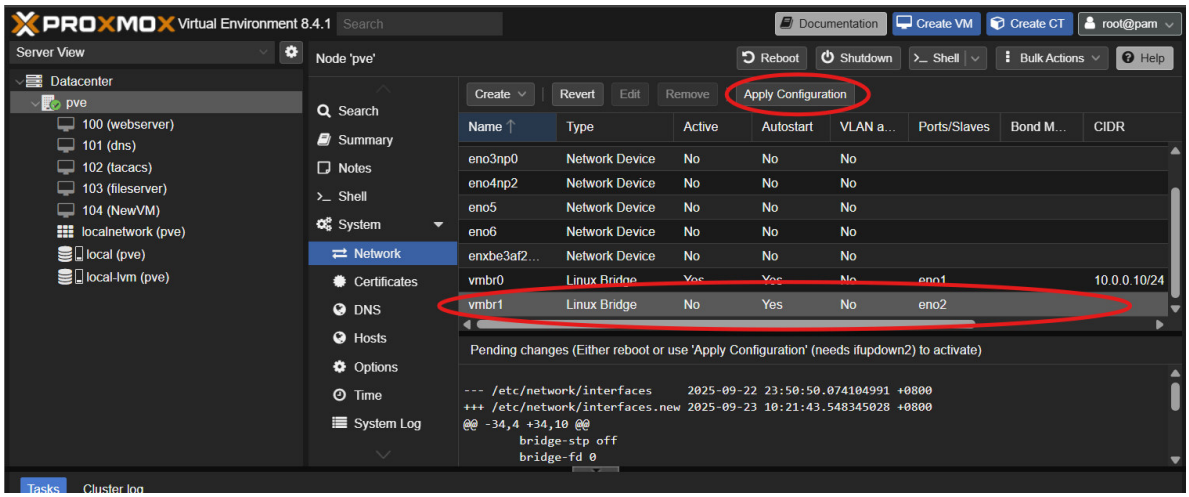


Figure Connect the Virtual Machine to the Network Switch.21

3.6 If the system asks for confirmation, please click **Yes** to apply the network changes.

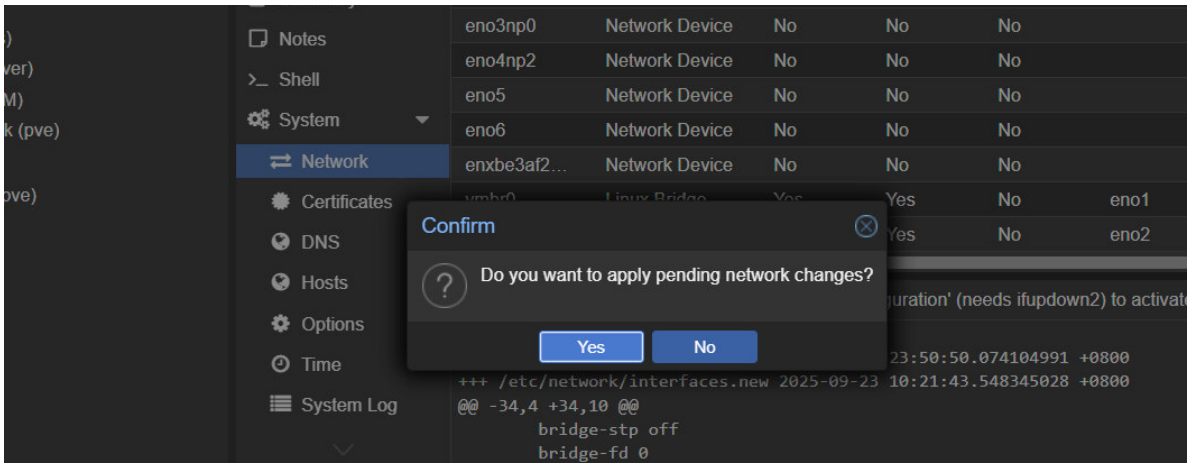


Figure Connect the Virtual Machine to the Network Switch.22

3.7 To allow the virtual machine to use the Linux Bridge, the Linux Bridge needs to be assigned to the virtual machine. Under the left panel, choose the VM that you wish to assign the Linux bridge to, and click on it. After clicking on the VM, in the right panel, click on the **"Hardware"**, select the **"Network Device"**, and click on the **Edit** button to make changes.

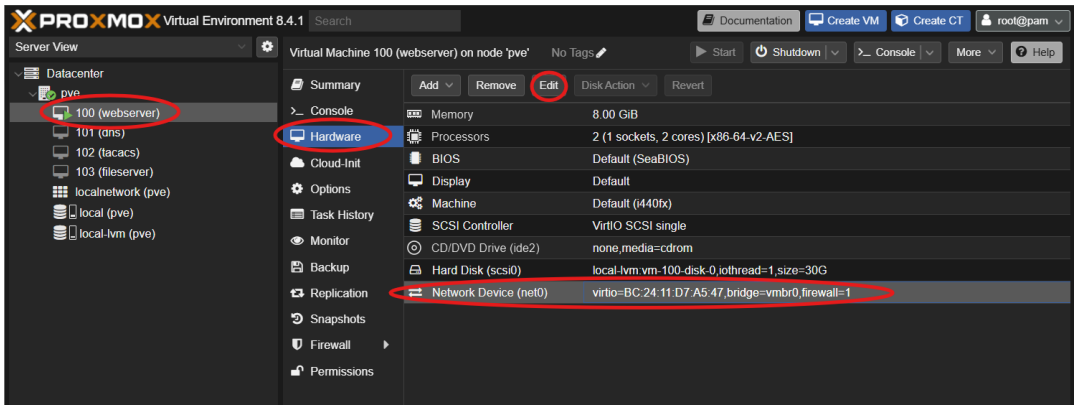


Figure Connect the Virtual Machine to the Network Switch.23

3.8 In the **Edit: Network Device** prompt, under the bridge dropdown selector, choose the new Linux bridge (for example, **vmbr1**) that was created in the last steps.

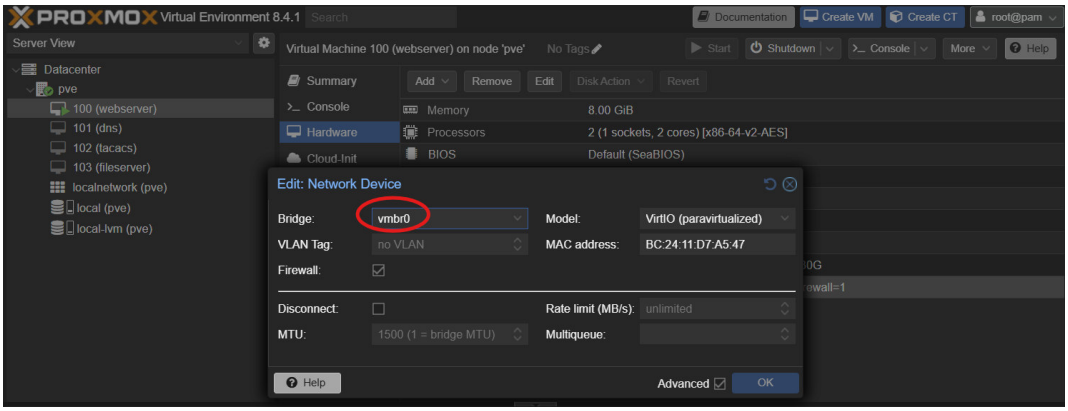


Figure Connect the Virtual Machine to the Network Switch.24

3.9 After choosing "vmbr1", click **OK** to confirm the changes.

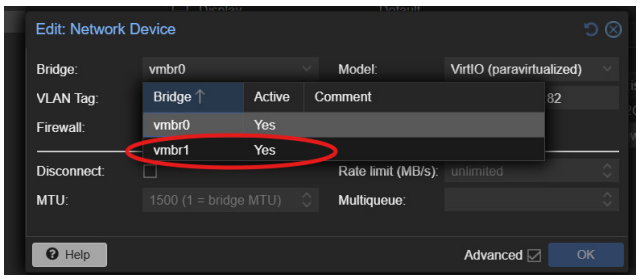


Figure Connect the Virtual Machine to the Network Switch.25

3.10 In the hardware page, the **network device** should change from **vmbr0** to **vmbr1**.

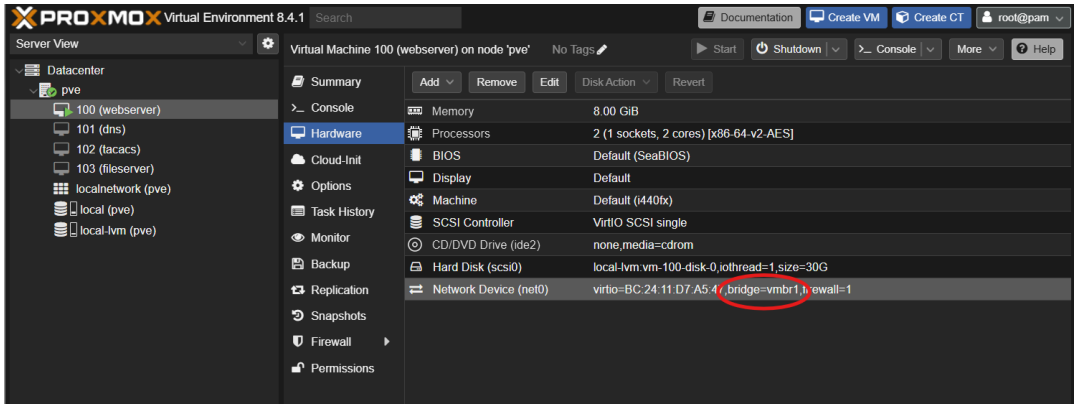


Figure Connect the Virtual Machine to the Network Switch.26

3.11 In the VM, power on (if you have not) and test if the VM can access the external network.

PART 4: Stop and Roll Back Virtual Machine

In this part, we will learn best practices on how to stop and roll back to the "fresh" state to avoid any issues when there are too many configurations made and causing the VM to crash or stop working.

Stopping a Virtual Machine

To prevent unnecessary corruption of the operating system in the VMs or the physical server, gracefully stop the virtual machine and Proxmox Server before powering off the rack.

4.1 To gracefully stop the virtual machine, run the command **"sudo shutdown now"** or click on the **"Shutdown"** button located in the respective Virtual Machine page.

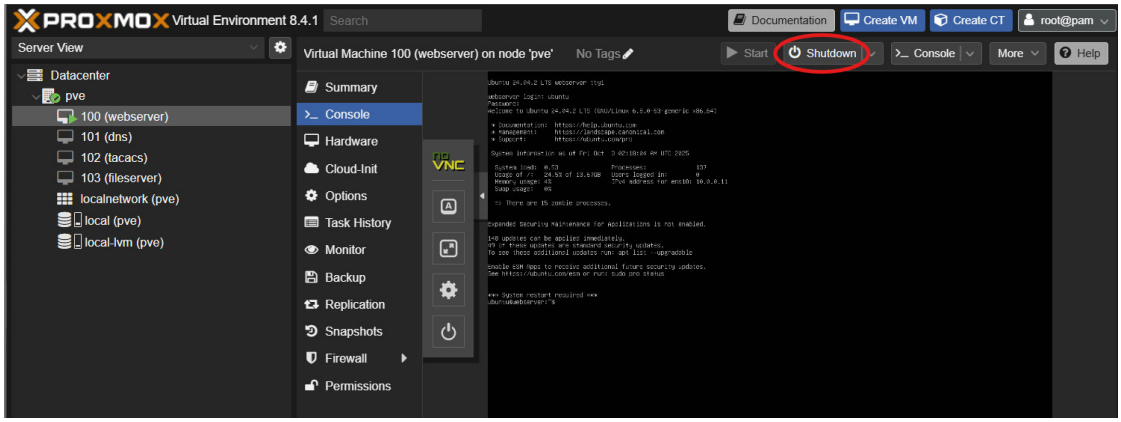


Figure Stop and Roll Back Virtual Machine.27

4.2 However, if the dropdown arrow is clicked, there will be several other options. **Reboot** – gracefully restart the VM, **Pause** - Suspends CPU execution but keeps the VM's RAM in memory, **Hibernates** – Saves the VM's entire memory state to disk and powers it off, **Stop** – Forcefully/immediately powers off the VM (not recommended), **Reset** – Force restarts the VM without shutting down cleanly (not recommended). It is always best practice to perform a graceful shutdown and graceful reboot of the VM, other than the rest of the options listed here.

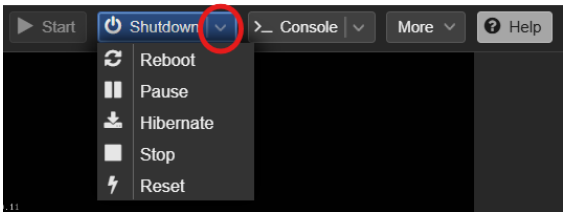


Figure Stop and Roll Back Virtual Machine.28

Roll Back a Virtual Machine

In each pre-configured virtual machine, there is one original snapshot image taken for each of them, and ***please do not delete them!!!***

You may roll back to the original snapshot image if you could not log in to any of the existing images due to some misconfiguration or software crash.

4.3 To check the original snapshot image, please go to the VM list, click on one of the VM names, for example, **100 (webserver)**, then choose **Snapshots**, and check the snapshot image with the name **"Original-Image".** The snapshot description indicates the default username and password to be used to log in to the default VM image snapshot.

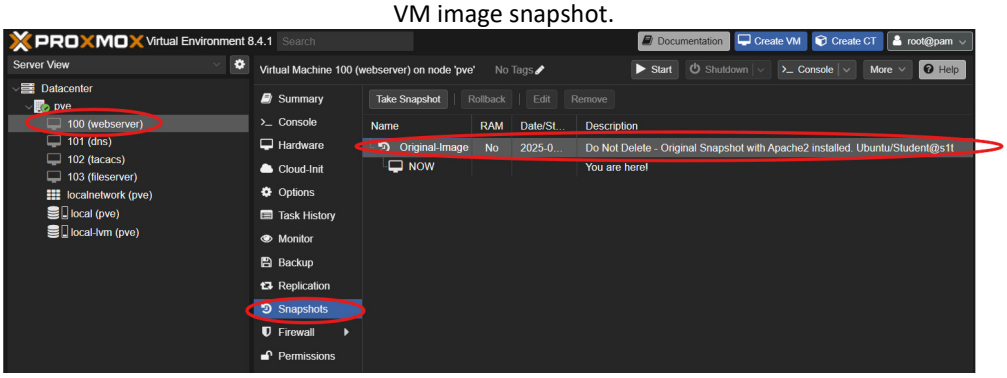


Figure Stop and Roll Back Virtual Machine.29

4.4 To roll back to the original snapshot of the specific virtual machine, go to the "**Original-Image"** name and then click on the **Rollback** button.

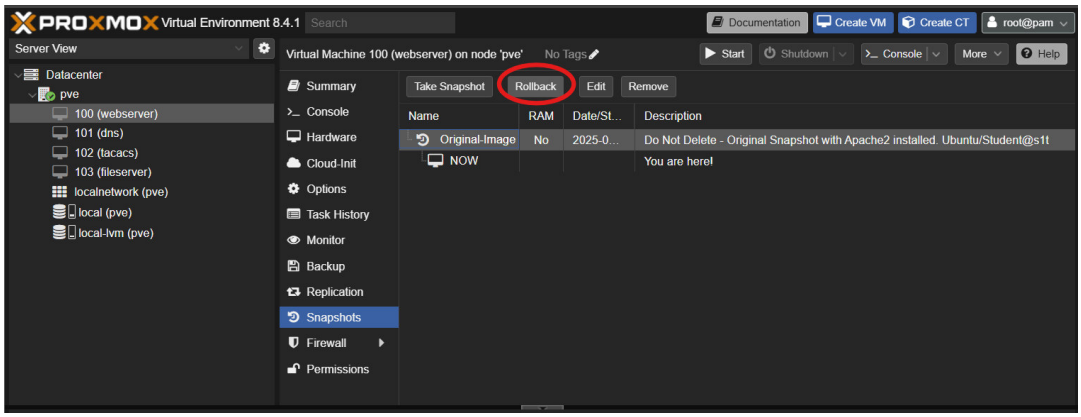


Figure Stop and Roll Back Virtual Machine.30

4.5 Click Yes to confirm the rollback.

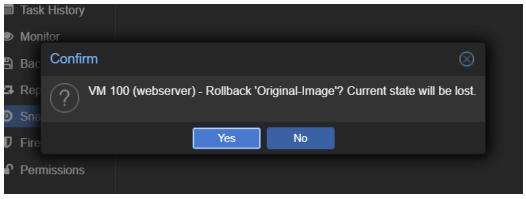


Figure Stop and Roll Back Virtual Machine.31

4.6 It will take some time to finish, and the virtual machine will be reverted with the original snapshot image with the default username and password.

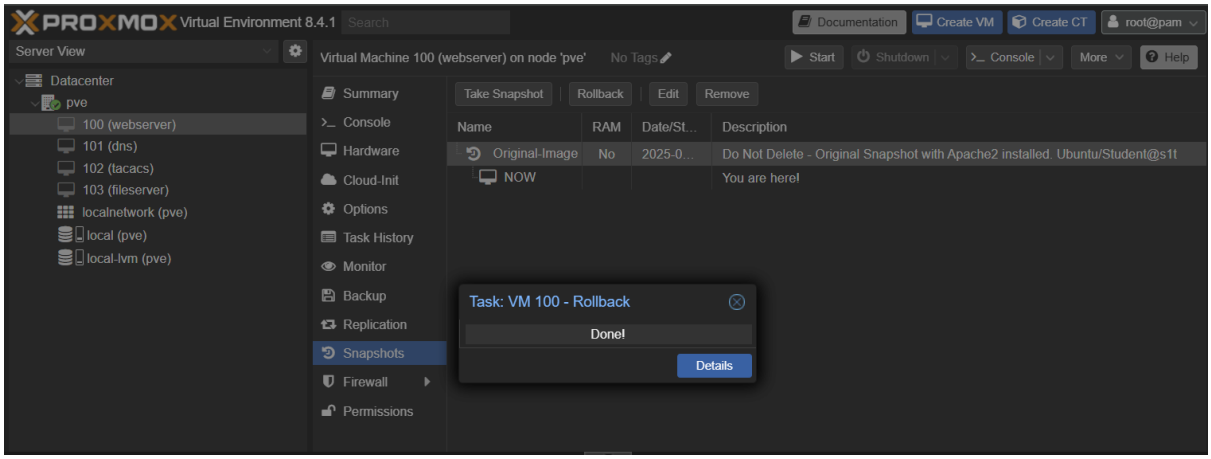


Figure Stop and Roll Back Virtual Machine.32

PART 5: Shutting down Proxmox Server

Graceful shutdown of Proxmox Server has very similar benefits to Virtual Machines' graceful shutdown. It also prevents unnecessary corruption of the operating system (Proxmox OS) before powering off the rack.

5.1 To gracefully stop the Proxmox Server, ensure all the virtual machines are no longer running, click on the **pve** host, and click on the "Shutdown" button on the top right.
You can also use the command **"shutdown now"** from the **">_Shell"** tab.

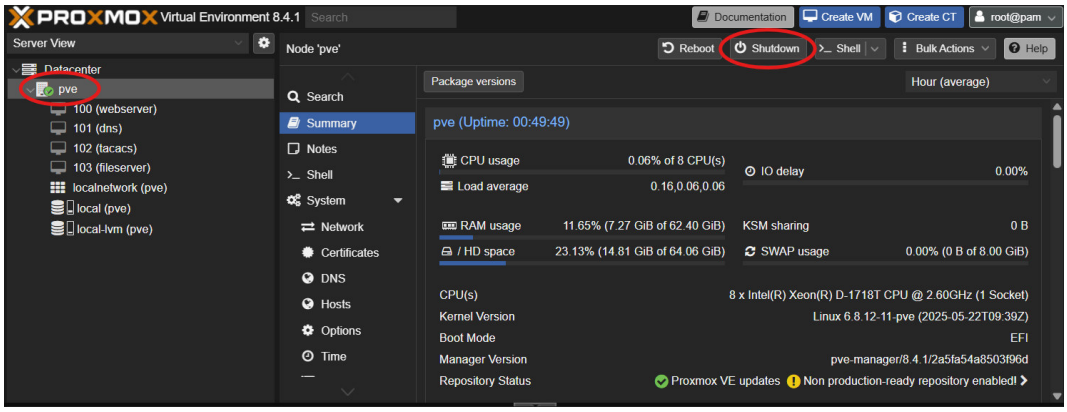


Figure Shutting down Proxmox Server.33

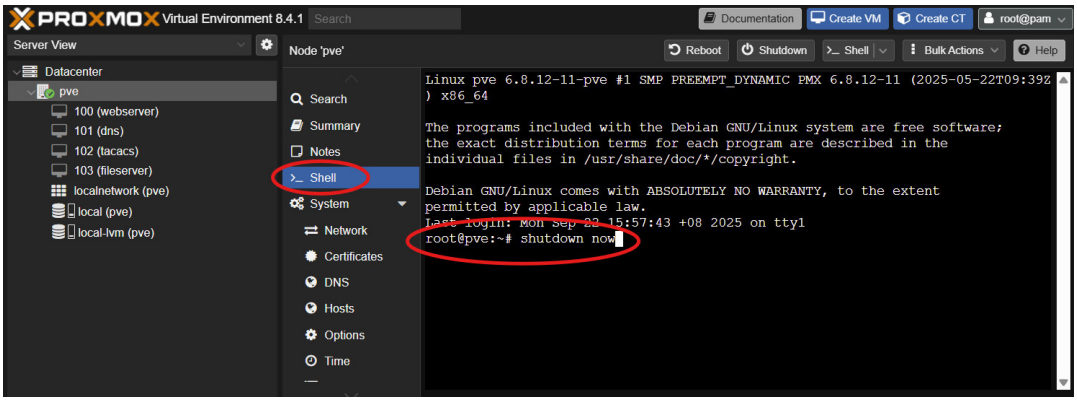


Figure Shutting down Proxmox Server.34

5.2 You can verify the server had been powered off by the remote console page or by no longer getting a feel on the breeze produced from the fan outlet located on the slot grill.



Figure Shutting down Proxmox Server.35

5.3 Congratulations! You have completed the lab!